
Reference Manual

Release 10.5

The Sage Development Team

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Here you find documentation for all of [Sage's](#) features, illustrated with lots of examples. A thematic index follows.
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USER INTERFACES

- Command Line Interface
- Jupyter Notebook Interface

GRAPHICS

- 2D Graphics
- 3D Graphics

3.1 Parents and Categories

- Parents and Elements
- Coercion
- Categories

3.2 Basic Rings and Fields

- Integers and Rational Numbers
- Real and Complex Numbers
- Commutative Polynomials
- Power Series and Laurent Series
- Finite Rings and Fields
- p -adic Numbers
- Noncommutative Polynomials
- Quaternion Algebras

3.3 Linear Algebra

- Matrices and Spaces of Matrices
- Vectors and Modules
- Tensors on Free Modules of Finite Rank

3.4 Calculus and Analysis

- Symbolic Calculus
- Mathematical Constants
- Elementary and Special Functions
- Asymptotic Expansions
- Numerical Optimization

3.5 Probability and Statistics

- Probability
- Statistics

3.6 Mathematical Structures

- Sets
- Monoids
- Groups
- Semirings
- Rings
- Algebras

3.7 Discrete Mathematics

- Combinatorics
- Graph Theory
- Quivers
- Matroid Theory
- Discrete Dynamics
- Coding Theory
- Cryptography
- Game Theory
- Symbolic Logic
- SAT solvers

3.8 Geometry and Topology

- Euclidean Spaces and Vector Calculus
- Combinatorial and Discrete Geometry
- Cell Complexes, Simplicial Complexes, and Simplicial Sets
- Manifolds and Differential Geometry
- Hyperbolic Geometry
- Parametrized Surfaces
- Knot Theory

3.9 Homological Algebra

- Chain Complexes and their Homology
- Resolutions

3.10 Number Fields, Function Fields, and Valuations

- Number Fields
- Function Fields
- Discrete Valuations
- Drinfeld Modules

3.11 Number Theory

- Diophantine Approximation
- Quadratic Forms
- L -Functions
- Arithmetic Subgroups of $SL_2(\mathbf{Z})$
- General Hecke Algebras and Hecke Modules
- Modular Forms
- Modular Symbols
- Modular Abelian Varieties

3.12 Algebraic and Arithmetic Geometry

- Schemes
- Plane and Space Curves
- Elliptic and Hyperelliptic Curves

3.13 Miscellaneous

- Databases
- Games

INFRASTRUCTURE

4.1 Programming Facilities

- Data Structures
- Utilities
- Test Framework
- Parallel Computing
- Python Technicalities

4.2 Subsystem Interfaces

- Interpreter Interfaces
- C/C++ Library Interfaces

4.3 Documentation System

- Documentation System

GENERAL INFORMATION

- External Packages
- Bibliographic References
- History and License

INDICES AND TABLES

- `genindex`
- `modindex`
- `search`